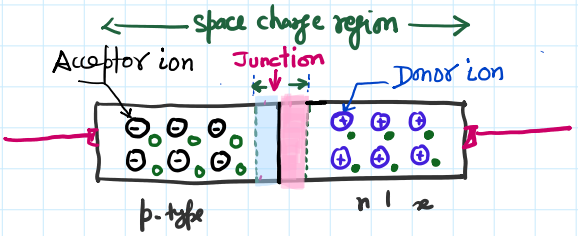
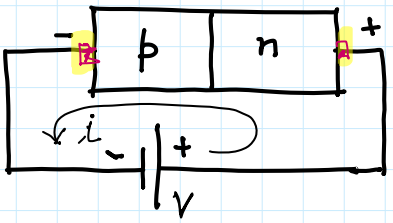
Lecture 27: p-n Junction Diode and its Characteristics

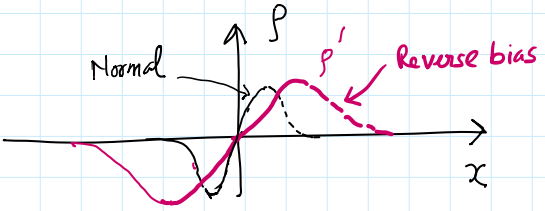
p-n Junction as Rectifier:



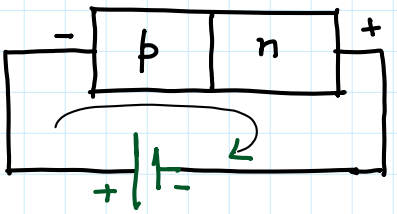
Flow of electrons is opposite to the direction of current hence are moving away from junction.

Flow of holes is in same direction to that of current and hence are moving away from junction.



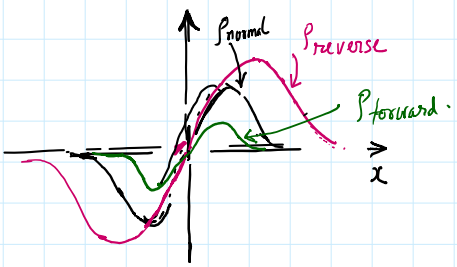


This increases the spread of charge density around the junction, thus making potential barrier high and stopping the condition, even if large voltage is applied. This is known as Reversed Biased Diode. OR Blocked Diode. Non-Conduction Mode.

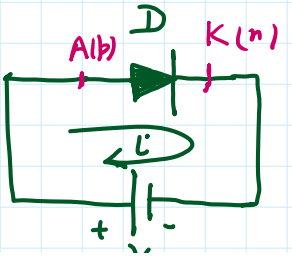


Forward Biased Diode:

With the shown connection of battery, the holes move to n-type region and electrons move to p-type region. The space charge density reduces with less potential barrier. This mode of Operation is referred as Forward Biased Diode OR Conduction mode

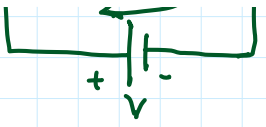


$V_{AK} \geq V_r$ (Forward biased)
Diode as 'ON' mode
$V_{AK} < 0$ (Reverse biased, OFF state)



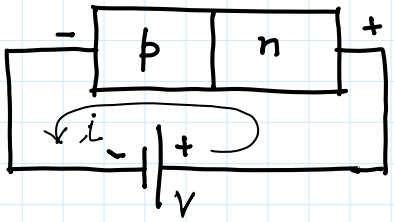
A = Anode
K = Cathode

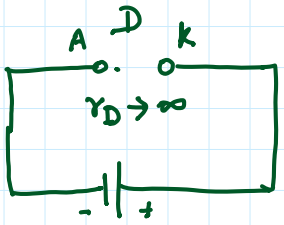
$V_{AK} < 0$ (Reverse Biased)



p-n Junction Diode:

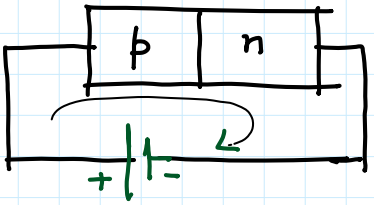
Reverse Biased

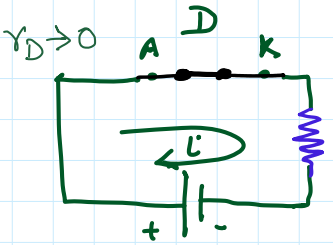




Current flows due to minority carriers

Forward Biased





Current flows due to majority carriers.

Remember:

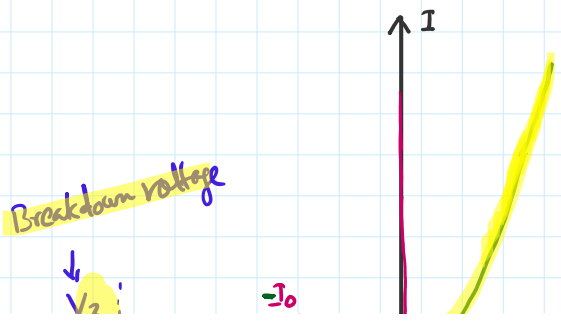
Region n-type: Electrons are majority carrier
Holes are minority carrier

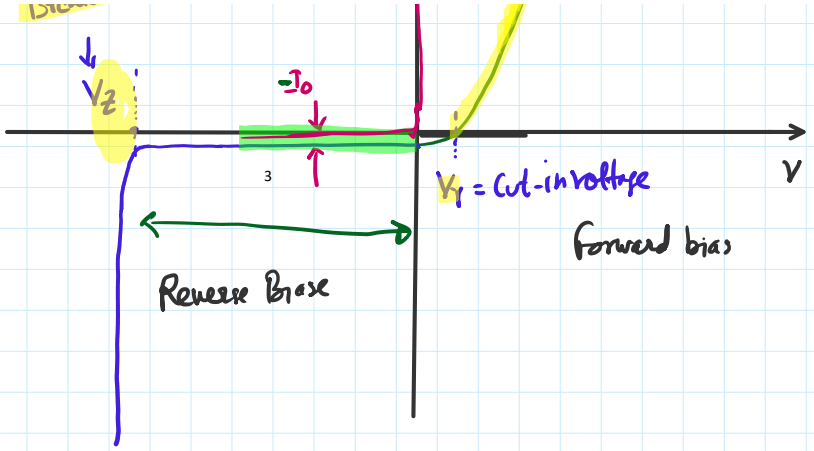
Region p-type: Holes are majority carriers.
Electrons are minority carriers

Forward current is produced by both types of majority carriers

Reversed current is produced by both types of minority carriers.

VI Characteristics of Diode





I_0 = Current due to minority carrier

$$I = I_0 \left(e^{\frac{(V-V_\gamma)}{\eta V_T}} - 1\right) ; \quad I_0 = \text{Current due minority carriers}$$

V = Applied voltage (V)

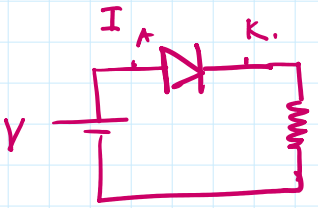
$$V_T = \frac{T}{11{,}600} ; \qquad T = 27^\circ C$$

$$T = 27 + 273 = 300^\circ K$$

$$= \frac{300}{11{,}600} = 0.026\ V = 26mV$$

$\eta = 1$ for Ge

$= 2$ for Si



(A) Diode forward biased: V is +ve

$$I = I_0 \left(e^{\text{+ve value}} - 1\right)$$

$$\boxed{I >> I_0}$$

(B) Diode Reverse Biased (V is large negative)

$$I = I_0 \left(e^{-\text{large value}} - 1\right) \simeq -I_0$$

In the following V_γ has been ignored, as it is quite small compared to V.

$$I = I_0 \left(e^{\frac{V}{\eta V_T}} - 1\right) \Rightarrow \left(\frac{I}{I_0} + 1\right) e^{\frac{V}{\eta V_T}}$$

$$\frac{dI}{\quad} = I_0 \boxed{e^{\frac{V}{\eta V_T}}} * \frac{1}{\eta V_T}$$

$$\frac{dI}{dV} = I_0 \left[e^{\frac{V}{\eta V_T}} \right] * \frac{1}{\eta V_T}$$

$$= \cancel{I_0} * \frac{(I+I_0)}{\cancel{I_0}} * \frac{1}{\eta V_T} = \frac{I+I_0}{\eta V_T}$$

$$\boxed{\frac{dV}{dI} = r_D = \frac{\eta V_T}{I+I_0}}$$

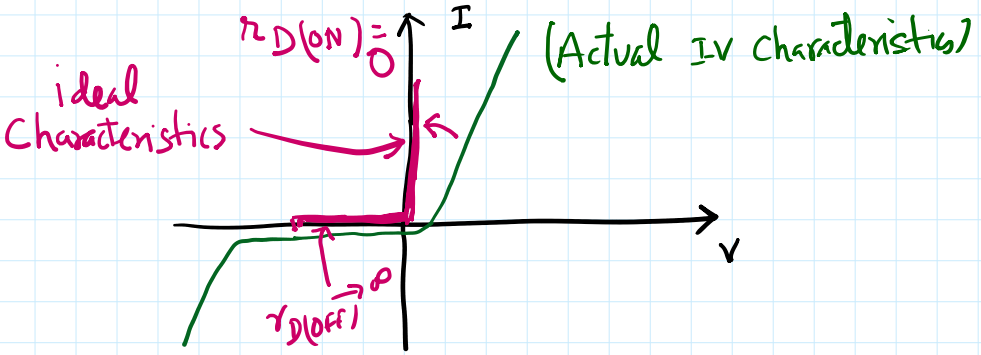
Forward biased (ON) mode: $I \gg I_0$

$r_D \to 0$, 1 ohm

Reverse biased (OFF) mode: $I \simeq -I_0$

$r_{D(OFF)} \to \infty$ (100 kΩ, 1 MΩ ...)

$r_{D(ON)} \simeq 0$



Types and Applications of Diode

Types of Diodes:





- **P-N Junction Diode**:
  The most fundamental type, this diode is made from semiconductor materials (p-type and n-type) and serves as a basic rectifier, allowing current flow in one direction.
- **Zener Diode**:
  Designed to operate in the reverse bias breakdown region, this diode maintains a constant voltage across its terminals, making it useful for voltage regulation.
- **Schottky Diode**:
  Known for its fast switching speed and low forward voltage drop, it's commonly used in high-frequency and power applications.
- **Light Emitting Diode (LED)**:
  When forward biased, an LED emits light, making it a ubiquitous choice for illumination and displays.
- **Photodiode**:
  This diode functions in reverse bias, generating a current when exposed to light, which allows it to detect and measure light.
- **Varactor Diode**:
  A voltage-controlled capacitor whose capacitance varies with the applied reverse voltage, used in tuning circuits.
- **Laser Diode**:
  Similar to an LED but designed to produce coherent laser light, used in optical devices and communication.
- **Rectifier Diode**:
  A type of diode specifically designed to convert alternating current (AC) to direct current (DC), often a high-power P-N junction diode.
- **Gunn Diode**:
  A specialized diode that exhibits negative differential resistance, useful in microwave oscillators and high-frequency circuits.
- **Avalanche Diode**:
  Operates by using avalanche breakdown to generate a large current, sometimes used in high-frequency oscillations or as a voltage regulator.
- **PIN Diode** is a semiconductor diode containing three layers: $\boxed{P^+ - I - N^+}$
  Here, I means a nearly intrinsic or very lightly doped semiconductor layer placed between the heavily doped p-type and n-type regions to withstand a higher reverse voltage and substantial current, woks effectively at radio and microwave frequencies

Solar Cell (Photovoltaic Diode):
A large-area P–N junction device that converts sunlight directly into electrical energy through the photovoltaic effect. Unlike a photodiode used for light detection, a solar cell normally operates without an external reverse-bias supply and delivers electrical power to a load.

Diode Circuit as Half Wave Rectifier.

Rectification of ac quantities is one of most common application of pn-junction diode.
A half wave rectifier circuit is shown in figure.

Operation: When positive voltage is across diode
∴ the diode conducts and current flows depending upon the value of R.

flows depending upon the value of R.

When negative voltage appears across diode is in non-conduction mode due to its reverse biasing. This stops current flowing as shown in the figure.